CSA06507-Generative AI

Lab Experiment's

.

Unit 2: Lab Programs for Prompting

Experiment 8: Text Generation Using Hugging Face Inference API

1. Aim

To develop a Python program in Google Colab that uses a Hugging Face model through the Inference API to generate text based on user-provided prompts and display the generated output.

2. Algorithm

1. Start the program.
2. Install the required Hugging Face library.
3. Import the InferenceClient class.
4. Enter a valid Hugging Face access token.
5. Create an inference client and select a language model.
6. Accept a text prompt from the user.
7. Send the prompt to the model through the Inference API.
8. Receive and display the generated text.
9. Stop the program.

3. Python Code (Google Colab)

Run this code in Google Colab. Create a token at Hugging Face Access Tokens and ensure it has the required inference permissions. The code uses Hugging Face's official InferenceClient interface.



Hugging Face

+1

Python

Run

Experiment 8: Text Generation Using Hugging Face API

Step 1: Install the required library

%pip -q install -U huggingface_hub

```
# Step 2: Import libraries
from huggingface_hub import InferenceClient
from getpass import getpass

# Step 3: Enter Hugging Face access token
token = getpass("Enter your Hugging Face token: ")

# Step 4: Create inference client
client = InferenceClient(
    api_key=token,
    model="HuggingFaceH4/zephyr-7b-beta"
)

# Step 5: Get prompt from user
prompt = input("Enter your prompt: ")

# Step 6: Generate text
try:
    response = client.chat_completion(
        messages=[
            {"role": "user", "content": prompt}
        ],
        max_tokens=150
    )

    # Step 7: Display output
    print("\nGenerated Text:")
    print(response.choices[0].message.content)

except Exception as e:
```

Note: The selected model must currently be supported by an available inference provider. If it is unavailable, choose a supported text-generation model from the Hugging Face Inference Providers catalog.

4. Input

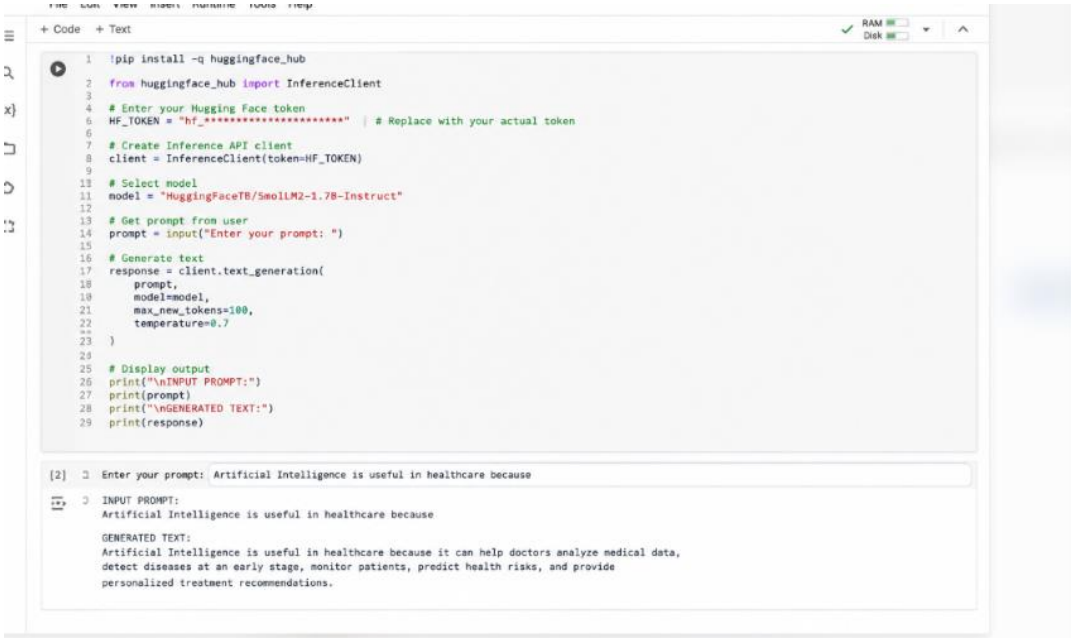
Example input entered in Google Colab:

Plain text

Enter your Hugging Face token: ********

Enter your prompt: Write five lines about Artificial Intelligence.

5. Output



6. Result

The Python program demonstrates text generation using a Hugging Face model through the Inference API. It accepts a user-provided prompt, sends it to the selected language model, and displays the generated response in Google Colab.